Zehua Ma

Generative AI | 3D Vision | Video Reasoning

mazh58@mail2.sysu.edu.cn | smazehua.github.io | github.com/smazehua

RESEARCH INTERESTS

Personalized generation, 3D generation and reconstruction, and video generation for visual reasoning. My work focuses on identity-preserving synthesis, faithful 3D geometry and textures, and task-relevant latent state trajectories for logically consistent video generation.

EDUCATION

Sun Yat-sen University

Sep. 2024 - Present; expected graduation 2027 | M.S. Student

- School of Intelligent Systems Engineering. Advisor: Prof. Xiaodan Liang.

Sun Yat-sen University

Sep. 2020 - Jun. 2024 | Undergraduate Studies

- School of Intelligent Systems Engineering. Advisor: Prof. Xiaodan Liang.

SELECTED RESEARCH PROJECTS

VR-JEPA: Learning Contrastive-State Latent Guidance for Generation-based Video Reasoning

2026 | Co-first author | arXiv:2609.40129

- Decouple state evolution reasoning from visual synthesis; predict task-relevant latent state trajectories in the V-JEPA 2.1 representation space to guide Wan2.2 video generation.
- Design localized contrastive-state learning to identify informative states and regions from successful and generated trajectories.
- Develop skill-routed mixture-of-experts with seven skill-specific experts and sparse routing for spatial, numerical, and physical reasoning.

ERGO: Excess-Risk-Guided Optimization for High-Fidelity Monocular 3D Gaussian Splatting

2026 | Co-first author | arXiv:2602.10278

- Use excess-risk decomposition for adaptive multi-view optimization in single-image 3D generation.
- Combine geometry-aware and texture-aware objectives to improve local reconstruction quality.
- Evaluate geometric fidelity and texture quality on Google Scanned Objects and OmniObject3D.

DreamVTON: Customizing 3D Virtual Try-on with Personalized Diffusion Models

2024 | Co-author | ACM Multimedia 2024

- Customize 3D human virtual try-on using personalized diffusion priors and multi-concept LoRA.
- Combine normal-style LoRA and template-based optimization to improve geometry and texture learning.

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Research experience & publications

INDUSTRY RESEARCH EXPERIENCE

Huawei | 2012 Laboratories

Nov. 2025 - Oct. 2026 | Research Intern

- Research visual reasoning through video generation; related work: VR-JEPA.
- Work on video generation and reasoning training pipelines, mixture-of-experts models, and contrastive learning.

Tencent | TEG, Hunyuan 3D Generation, Shenzhen

Jun. 2024 - Dec. 2024 | Research Intern

- Study single-image 3D reconstruction and the impact of synthetic multi-view data on LRM generalization and detail recovery.
- Build synthetic multi-view data workflows, fine-tune LRMs, and train local quality assessment models.

Tencent | TEG, Hunyuan Image Generation, Shenzhen

Nov. 2023 - Jun. 2024 | Research Intern

- Work on identity-preserving portrait generation for personalized ID photos.
- Train face-identity and template LoRAs and combine them to preserve identity and control appearance.

ZMO.AI | Shenzhen

Jul. 2023 - Nov. 2023 | Computer Vision Algorithm Intern

- Generate product advertising images while preserving the product across different scenes.
- Combine product segmentation, layout composition, background generation, and local inpainting to improve boundaries, lighting, and contact shadows.

ADDITIONAL CO-AUTHORED PUBLICATIONS & PREPRINTS

Beyond the Eye: Efficient Multimodal Reasoning via Self-Regulated Implicit Visual Tools.
Xiuwei Chen, Quanlin Chen, Wentao Hu, Zisheng Chen, Kun Xiang, **Zehua Ma**, et al.
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI). Accepted October 2026.
Preprint: arXiv:2607.11106.

AnyCrowd: Instance-Isolated Identity-Pose Binding for Arbitrary Multi-Character Animation.
Zhenyu Xie, Ji Xia, Michael Kampffmeyer, Panwen Hu, **Zehua Ma**, et al.
arXiv:2603.15415, 2026.

METHODS & TOOLS

Diffusion models, personalized generation, LoRA, DreamBooth, ControlNet, inpainting, 3D Gaussian Splatting, large reconstruction models (LRMs), JEPA, mixture-of-experts, and contrastive learning.